

A Project Report on

ACTIVE CELL BALANCING OF LITHIUM-ION BATTERIES FOR AN EV

Submitted in partial fulfillment of the requirement for the degree of

Bachelor of Technology in

Electronics and Instrumentation Engineering

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This is to certify that Mr. **DANNANA THARUN (228867602011)**, a final year B.Tech. student in the Department of Electronics and Instrumentation, AKNU college of Engineering, Adikavi Nannaya University, has successfully completed a project titled "**ACTIVE CELL BALANCING OF LITHION ION BATTERIES FOR AN EV**" at the **Indian Institute of Petroleum and Energy (IIPE), Visakhapatnam**, under my supervision from **08-12-2025** to **07-04-2026**

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DECLARATION

DANNANA THARUN (228867602003) , KOTHA DEVI PRASAD (228867602011) hereby declare that the project work report in the filed “**ACTIVE CELL BALANCING OF LITHION ION BATTERIES FOR AN EV**” done by me under the guidance of Dr. D. Kishore, Assistant Prof., Department of Electronics and Instrumentation Engineering, Adikavi Nannaya University College of Engineering, Adikavi Nannaya University, is submitted for partial fulfilment of requirement for the award of Bachelor of Technology in Electronics and Instrumentation Engineering in the Academic year 2025-2026.

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CONTENTS

I. ABSTRACT.....	6-6
II. INTRODUCTION.....	6-6
1. Battery Management system.....	6-7
1.1.1. Rechargeable batteries.....	7-8
1.1.2. Parameters of BMS.....	8-9
1.1.3. voltage, current, temperature	9-9
1.2. States estimations.....	10-10
1.2.1. State of charge	10-10
1.2.2. state of health	10-10
1.3. Charging algorithm.....	10-10
1.4. Cell balancing.....	11-11
1.5. Communication.....	11-11
III. METHODOLOGY	
IMPLIMENTING DIFFERENT BMS MODELS IN MATLAB	
A. Soc estimation.....	12-15
a. Coulomb counting method	12-12
b. Kalman filter method.....	12-15
c. SoH estimation	15-16
d. Charging of electric vehicle.....	17-18
e. Cell balancing	
i. Passive cell balancing	18-19
ii. Active cell balancing	19-40
➤ Capacitor based balancing.....	23-26
➤ DC-to-DC converter-based balancing	26-40
1. battery pack realistic behavior for better active cell balancing	27-30
2. Soc estimation using Kalman filter for battery pack	30-31
3. Pi controller and pwm.....	32-33
4. Cell selection logic.....	33-34
5. Switching logic.....	35-37
6. Gate driver	37-38
7. Inductor based DC-DC converter.....	38-40
IV. RESULTS	41-42
V. CONCLUSION	42-42
VI. REFERENCES.....	43-44

ACTIVE CELL BALANCING OF LITHION ION BATTERIES FOR AN EV

I. ABSTRACT

Greenhouse gases and global warming are severe issues for the living on earth and the main factor of these issues is the emission of harmful gases from fossil fuel or petroleum-used vehicles. Thus, the electric vehicle is proposed and commercialized to cut down the use of fossil fuel vehicles. However, the problem of battery cell imbalance of the electric vehicle leads to unsatisfactory and critical bad impacts on the battery lifespan and operating duration, in turn, reducing the public interest in electric vehicles. We are trying to solve the cell unbalanced issue by utilizing the active cell balancing methods based on the Buck-Boost DC/DC Converter topologies with the integration of several closed-loop control systems and algorithms on the battery pack designed by battery builder app which battery pack is similar to the practical electric vehicle. MATLAB/Simulink is used for the development of this simulation model.

II. INTRODUCTION:

The increasing demand for sustainable transportation has led to rapid advancements in electric vehicle (EV) technology. Electric vehicles rely heavily on rechargeable battery systems as their primary energy source. Among the various battery technologies available, lithium-ion batteries have become the most widely used due to their high energy density, long cycle life, and efficient charge–discharge characteristics. However, the performance, safety, and lifespan of these batteries depend on proper monitoring and control systems.

1. BATTEY MANAGEMENT SYSTEM (BMS) :

A Battery Management System (BMS) is an electronic control system designed to monitor, manage, and protect rechargeable battery packs. It plays a crucial or an important role in systems that depend on batteries as the primary energy source, such as electric vehicles, renewable energy storage systems, consumer electronics, and industrial equipment. Modern battery technologies, especially lithium-ion batteries, provide high energy density and long cycle life, but they require strict control of operating conditions. Without proper monitoring and management, batteries can degrade quickly or operate unsafely. The

BMS ensures that the battery operates within safe limits while delivering optimal performance and extending the overall lifespan of the battery pack.

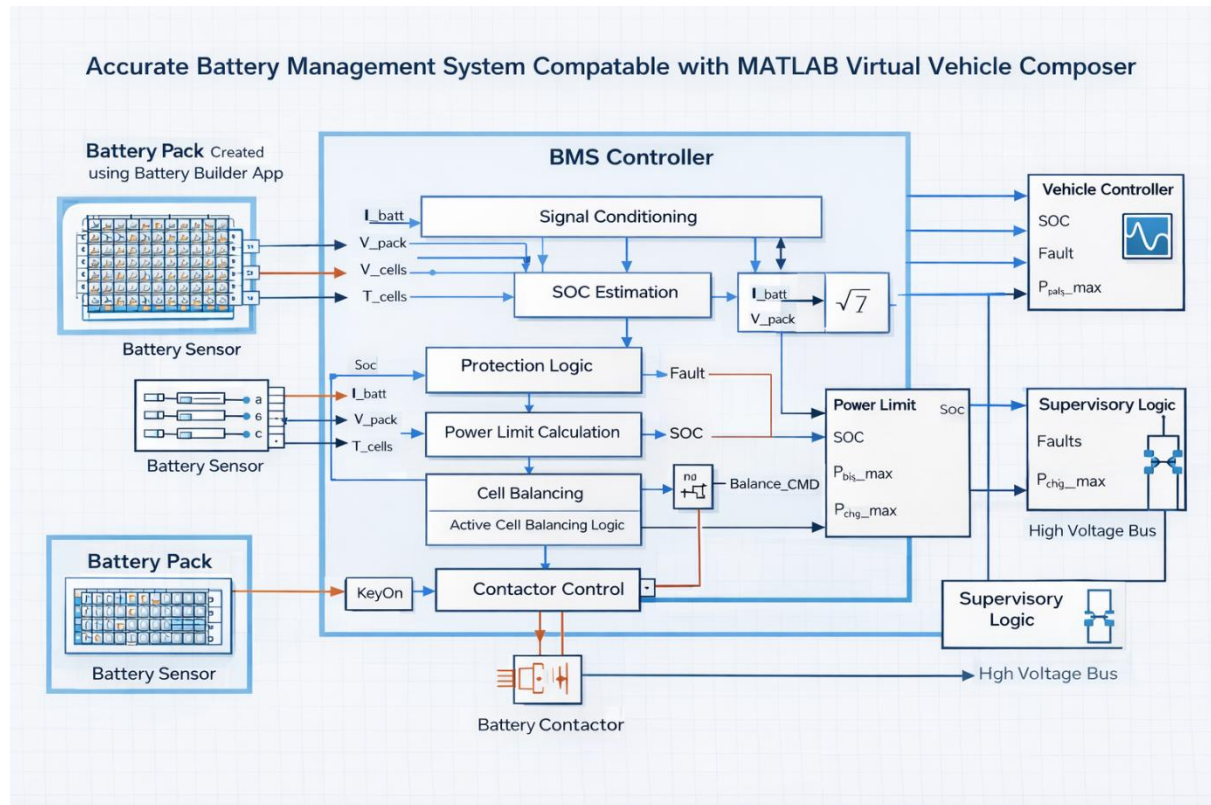


Figure 1 Model Battery Management system

The architecture of a typical BMS at **Figure 1**, includes several main components. These components include sensors, measurement circuits, microcontrollers, balancing circuits, and protection devices. Sensors measure parameters such as voltage, current, and temperature. Measurement circuits convert these signals into digital data that can be processed by the microcontroller. The microcontroller acts as the brain of the BMS, executing control algorithms and making decisions based on the measured data. Balancing circuits help equalize cell voltages, while protection devices ensure safe operation by disconnecting the battery when necessary.

1. 1.1. RECHARGEABLE BATTERIES

Rechargeable batteries consist of electrochemical cells that convert chemical energy into electrical energy during discharge and convert electrical energy back into chemical energy during charging. There are different types of batteries and sizes which are shown in the **Figure 2**. A single battery cell usually produces a relatively low voltage. For example, a typical lithium-ion cell provides about 3.6 to 3.7 volts **Figure 2**. To achieve the voltage and capacity required for large applications such as electric vehicles, many cells are connected together to form a battery pack. Cells connected in series increase the total voltage, while cells connected in parallel increase the total capacity and current capability. For instance, a battery pack configuration may include dozens or even hundreds of cells arranged in a series-parallel structure. Because of the

large number of cells involved, monitoring and controlling each cell becomes essential for safe operation. This is where the Battery Management System becomes necessary.

DIFFERENT TYPES OF BATTERIES



Figure 2 Different types batteries

The main purpose of a BMS is to supervise or manage the entire battery pack and maintain its operation within safe electrical and thermal limits. The BMS performs several important functions including measurement, monitoring, protection, state estimation, cell balancing, and communication. These functions work together to ensure that the battery pack operates efficiently and safely under different conditions such as charging, discharging, and standby operation.

1. 1.2. PARAMETERS OF THE BMS

One of the most fundamental tasks of the BMS is battery monitoring. Monitoring involves continuously measuring various parameters of the battery pack. The most important parameters are include voltage, current, and temperature. Voltage monitoring is essential because each battery cell must operate within a specific voltage range. If the voltage of a lithium-ion cell exceeds the maximum limit during charging, it may cause internal chemical reactions that lead to overheating, swelling, or even fire hazards. On the other hand, if the voltage drops below the minimum

limit during discharge, the battery may experience permanent capacity loss or internal damage. Therefore, the BMS continuously measures the voltage of each cell and the overall pack voltage to ensure safe operation.

1.1.3. CURRENT, VOLTAGE, TEMPERATURE

Current monitoring is another important function of the BMS. Current represents the flow of electrical charge into or out of the battery pack. During charging, current flows into the battery, while during discharging it flows out to supply energy to the load. Excessive current can cause overheating and accelerate battery degradation. To prevent this, the BMS uses current sensors such as shunt resistors or Hall-effect sensors to measure the current accurately. Based on the measured current values, the BMS can limit or interrupt charging and discharging if abnormal conditions are detected (Lim et al.; Paidi et al. 2022).

In a Battery Management System (BMS), voltage acts as the primary indicator for safety, capacity, and health by monitoring both individual cells and the entire pack. It ensures safety by triggering disconnects during overvoltage (to prevent fire) or undervoltage (to prevent cell damage), while also serving as a digital "fuel gauge" to estimate the State of Charge (SoC). Furthermore, the BMS uses voltage differentials to perform cell balancing, bleeding off energy from stronger cells so the entire pack stays uniform, and it tracks "voltage sag" under load to calculate internal resistance and determine the battery's overall State of Health (SoH).

Temperature monitoring is critical in the BMS because battery performance and safety are strongly affected by temperature. Lithium-ion batteries typically operate best within a moderate temperature range. High temperatures can accelerate chemical reactions within the battery and may lead to thermal runaway, a dangerous condition where the battery temperature rises uncontrollably. Extremely low temperatures can reduce battery performance and make charging unsafe. To manage temperature effectively, sensors are placed at different locations within the battery pack. The BMS collects temperature data and takes necessary actions they are activating cooling systems, limiting current flow, or shutting down the system when temperature limits are exceeded. (Lim et al.)

Thermal management is closely linked with the operation of the BMS. During battery operation, heat is generated due to internal resistance and electrochemical reactions. If this heat accumulates, it can affect battery performance and safety. Therefore, battery packs often include cooling systems such as air cooling or liquid cooling. The BMS monitors temperature sensors and controls the thermal management system to maintain safe operating temperatures

1.2. STATES ESTIMATIONS:

Another key function of the BMS is battery state estimation. Battery state estimation refers to the process of determining important internal parameters that cannot be measured directly. The most commonly estimated parameter is the State of Charge (SOC). SOC indicates the remaining energy available in the battery relative to its maximum capacity. It is usually expressed as a percentage, where 100 percent(%) means the battery is fully charged and 0 percent(%) means it is completely discharged. Accurate SOC estimation is important for predicting driving range in electric vehicles and for managing charging and discharging operations efficiently.

1. 2.1. STATE OF CHARGE (SOC)

SOC estimation can be performed using several methods. One common method is coulomb counting, which calculates SOC(Paidi et al. 2022) by measuring the amount of current flowing into or out of the battery over time. Another method involves using open-circuit voltage measurements and battery models. More advanced approaches combine mathematical models and estimation algorithms such as Kalman filtering to achieve higher accuracy. The BMS uses these methods to provide reliable SOC information to the system and the user.

1.2.2. STATE OF HEALTH(SOH)

In addition to SOC, the BMS may estimate the State of Health (SOH) of the battery. SOH represents the condition of the battery relative to its original state when it was new. As batteries age, their capacity gradually decreases and internal resistance increases. These changes affect the performance and efficiency of the battery pack. By analyzing voltage, current, temperature, and usage patterns, the BMS can estimate SOH and predict the remaining useful life of the battery. This information is valuable for maintenance planning and system reliability.

1.3. CHARGING ALGORITHM

the Constant Current-Constant Voltage (CC-CV) technique is the industry-standard approach for modeling battery charging. The process is typically implemented using Simscape Electrical components, where a Controlled Current Source initially maintains a fixed charging rate (CC phase) to rapidly increase the State of Charge (SOC). As the battery terminal voltage reaches its regulated limit, a control loop—often managed via State flow or PID Controllers—transitions the system to a Constant Voltage (CV) phase. During this stage, the charging current exponentially decays, allowing the cells to reach full capacity while preventing overvoltage. This dual-stage modeling is critical for accurately evaluating thermal behavior, charging efficiency, and the impact of fast-charging protocols on battery life cycles.

1.4. CELL BALANCING

Cell balancing(Paidi et al. 2022) is another essential feature of a Battery Management System. Because battery cells are not perfectly identical, their voltage levels and charge capacities can differ slightly. Over time, these differences can increase, causing some cells to become more charged than others. When the battery pack is charged, the cell with the highest voltage may reach its limit first, forcing the charging process to stop even though other cells are not fully charged. Similarly, during discharge, the weakest cell may reach the minimum voltage limit earlier than the others. This imbalance reduces the usable capacity and efficiency of the entire battery pack

To solve this problem, the BMS performs cell balancing. Cell balancing equalizes the charge levels of individual cells so that they operate more uniformly. There are two main types of balancing techniques: passive balancing and active balancing. Passive balancing is the simpler and more commonly used method. In this approach, excess energy from higher voltage cells is dissipated as heat through resistors. Active balancing, on the other hand, transfers energy from higher charged cells to lower charged cells using electronic circuits such as capacitors, inductors, or DC-DC converters. Active balancing is more efficient but also more complex and expensive.

Charging control is another critical function managed by the BMS. Lithium-ion batteries require a specific charging strategy to prevent damage and ensure long life. The most widely used method is the Constant Current–Constant Voltage (CC-CV) charging technique. During the constant current phase, the battery is charged with a fixed current until the cell voltage reaches the maximum allowable level. After reaching this voltage, the system switches to the constant voltage phase, where the voltage is maintained while the charging current gradually decreases. The BMS supervises this entire process and ensures that charging stops when the battery is fully charged

1. 5. COMMUNICATION

Another important aspect of modern Battery Management Systems is communication. In many applications, the BMS does not operate independently but interacts with other electronic systems. For example, in an electric vehicle, the BMS communicates with the vehicle control unit, motor controller, charger, and display system. Communication allows the BMS to share information about battery status, SOC, temperature, and fault conditions. This information helps the vehicle control system manage power flow and ensure safe operation. Communication protocols such as CAN (Controller Area Network) are commonly used for data exchange between electronic control units.

In modern engineering practice, BMS development often involves modeling and simulation before implementing the system in hardware. Simulation tools such as MATLAB and Simulink are widely used for this purpose. These tools allow engineers to create mathematical models of battery cells, simulate charging and discharging behavior, implement SOC estimation algorithms, and test control strategies. Simulation helps engineers evaluate system performance under different conditions without the risks associated with physical experiments.

III. METHODOLOGY

IMPLIMENTING DIFFERENT BMS MODELS IN MATLAB

B. SOC ESTIMATION

- a. Coulomb counting method
- b. Kalman filter method

a. COULOMB COUNTING METHOD

The Coulomb counting method is a widely used technique for estimating the State of Charge (SOC) of a battery by measuring and integrating the current flowing into and out of the battery over time. The fundamental principle is that the change in stored charge is directly proportional to the current, and by subtracting the discharged charge from the initial SOC, the remaining battery capacity can be determined. This method is commonly implemented in MATLAB and Simulink due to their strong capabilities in numerical computation and system modelling. The SOC at any instant is calculated based on the initial SOC, battery capacity, and the integral of current with respect to time, where discharging current decreases SOC and charging current increases it.

In practical implementation, MATLAB uses numerical integration techniques to compute SOC by updating its value at each time step using measured current data and known battery capacity. In Simulink, the design involves a current sensor connected to the battery model, followed by a gain block representing the inverse of capacity and an integrator block to estimate SOC continuously. This estimated SOC is essential for battery monitoring, control, and management applications such as charging control and active cell balancing in large battery packs. Despite its simplicity and ease of implementation, the Coulomb counting method requires accurate current sensing and a reliable initial SOC value, as errors can accumulate over time and affect estimation accuracy.

B. KALMAN FILTER METHOD

The Kalman Filter-based State of Charge (SOC) estimation method is an advanced and widely used technique in electric vehicle battery management systems due to its high accuracy and robustness against noise. In this work, the SOC estimator is implemented using MATLAB and Simulink, where a Kalman Filter block is integrated with the battery model. The estimator takes key inputs such as battery current, cell voltage, temperature, and initial SOC, as shown in the developed model at **Figure 3**. Unlike simple methods such as Coulomb counting, the Kalman Filter continuously corrects the SOC estimate by comparing the measured battery voltage with the predicted voltage obtained from the battery model. This makes it highly effective in reducing accumulated errors and improving estimation reliability under dynamic operating conditions.

In the developed Simulink model, the battery is represented using an equivalent circuit model, and sensors are used to measure real-time current and voltage. These signals are fed into the SOC Estimator (Kalman Filter) block, which internally uses a prediction-correction mechanism. During the prediction step, the filter estimates the SOC based on the previous state and input current, while in the correction step, it updates the SOC using the difference between measured and estimated voltage.

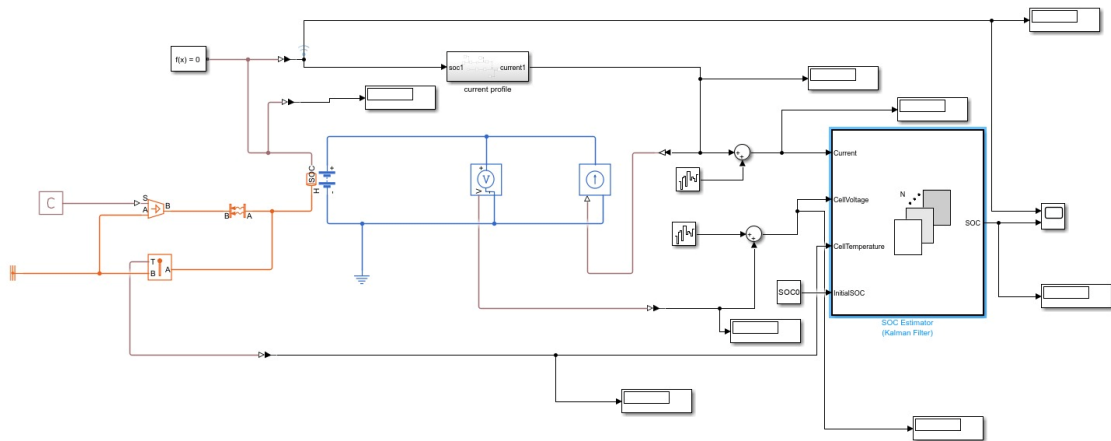


Figure 3. Simulink Model of SOC Estimator Using Kalman Filter.

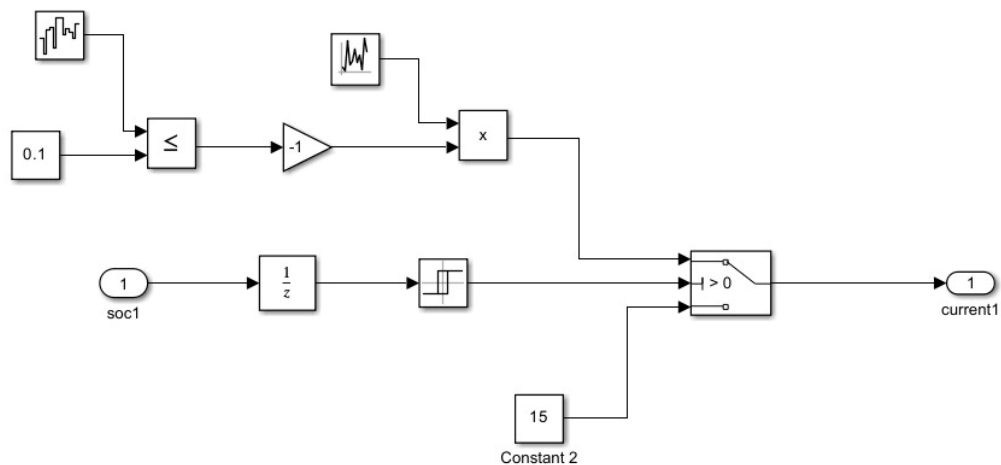


Figure 4 represents the current profile generation logic, where switching conditions and signal processing blocks are used to simulate realistic charge/discharge behaviour. This dynamic current profile is essential for testing the estimator under varying load conditions, ensuring that the Kalman Filter responds accurately to real-world scenarios.

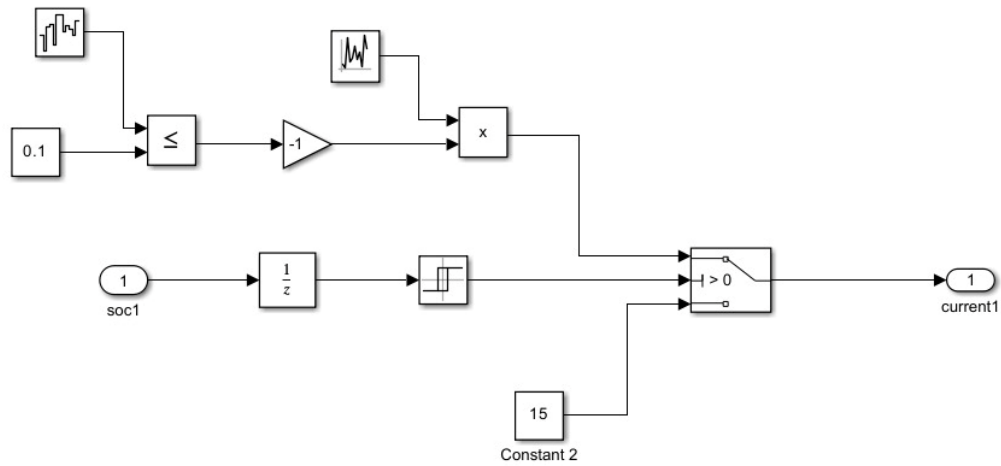


Figure 4. Current Profile of SOC Estimator.

Overall, the Kalman Filter-based SOC estimation model provides a more accurate and stable estimation compared to traditional methods, especially for large battery packs used in electric vehicles. It effectively handles system uncertainties, measurement noise, and nonlinear battery characteristics. The integration of this estimator with the battery model allows real-time monitoring of SOC, which is crucial for safe operation, performance optimization, and implementation of advanced strategies such as active cell balancing. Therefore, the developed

Table 1. MATLAB Library Tools

Block Name	Library / Type	Settings
MATLAB Function (EKF Core)	Simulink > User-Defined Functions > MATLAB Function	Implement state equation: $x=[SOC, V1, V2]$, predict + update steps. See code template below
Unit Delay (State Memory)	Simulink > Discrete > Unit Delay	Stores previous state vector $[SOC, V_{RC1}, V_{RC2}]$. IC = $[0.7; 0; 0]$ per cell
OCV Lookup Table	Simulink > Lookup Tables > 1-D Lookup Table	Map SOC (0-1) to OCV (2.7-4.2V). Use measured OCV-SOC curve data
For Each Subsystem	Simulink > Ports & Subsystems > For Each Subsystem	Wrap EKF to run independently on each of 900 cells
Mux (EKF inputs)	Simulink > Signal Routing > Mux	Combine Vcell, Icell, Tcell per iteration

Table 2. MATLAB Library Tools.

model demonstrates a practical and efficient approach for SOC estimation in EV battery packs, making it suitable for research and advanced battery management system design. There are some more important library tools and its settings are at above Table 1.

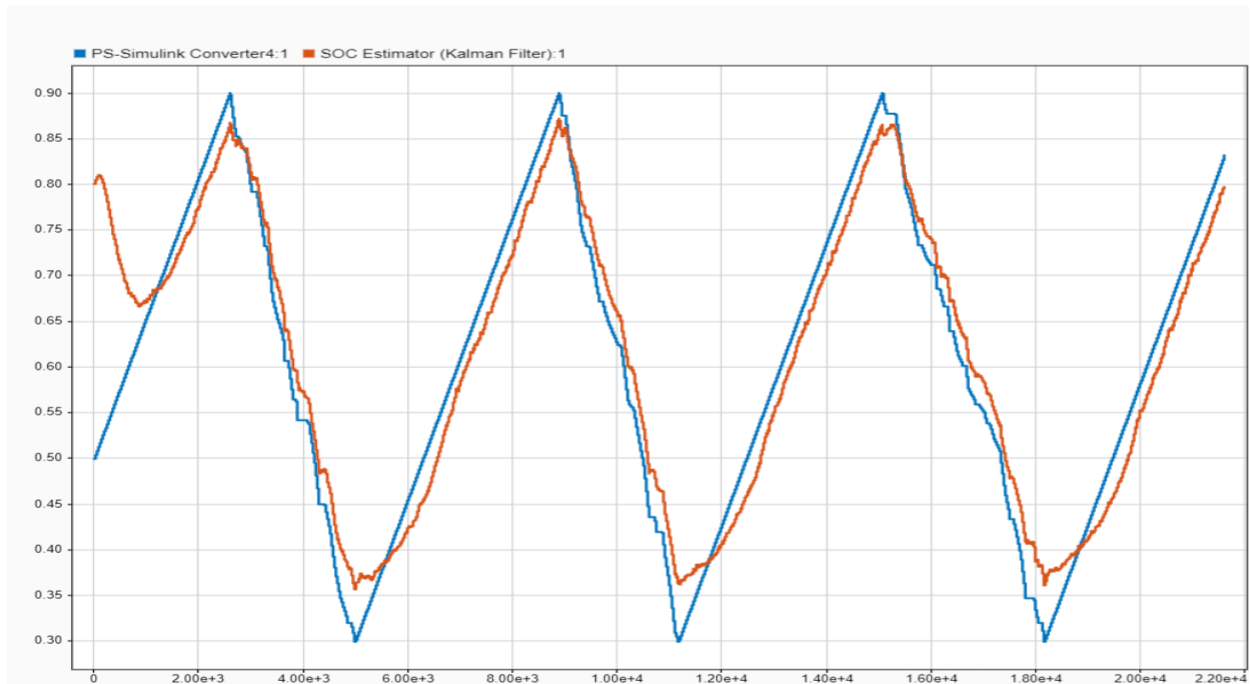


Figure 5.SOC Estimation Using Kalman Filter.

The output waveform in **Figure 5** shows the comparison between the **actual State of Charge (SOC)** and the **estimated SOC using the Kalman Filter** in your MATLAB Simulink model. The repeating rising and falling pattern indicates that the battery is undergoing cyclic charging and discharging based on the applied current profile. During charging intervals, the SOC increases, while during discharging it decreases, forming a periodic waveform. This confirms that your model is operating under dynamic conditions similar to real electric vehicle battery usage.

The two curves closely follow each other, which indicates that the Kalman Filter is accurately estimating the SOC. Small deviations between the actual and estimated SOC can be observed, especially during rapid transitions, due to factors such as measurement noise and model inaccuracies. However, the estimated SOC quickly converges back to the actual value, demonstrating the effectiveness of the Kalman Filter in correcting errors. Overall, the result shows that the developed SOC estimation model is reliable and suitable for practical battery management applications.

C. SOH ESTIMATION

The above model represents the **State of Health (SOH) estimation** of an EV battery pack using an adaptive Kalman Filter approach in MATLAB Simulink. SOH indicates the overall condition of the battery compared to its original state and is typically expressed as a percentage, where 100% represents a new battery and lower values indicate aging or degradation. In **Figure 6**, the SOH estimator block takes inputs such as estimated SOC, temperature, and internal resistance (R_0), which is also estimated by the SOC estimator. The

inclusion of sensor noise in current and voltage signals makes the model more realistic, reflecting practical operating conditions. The Kalman Filter continuously processes these inputs and updates the SOH by tracking changes in battery parameters, especially internal resistance, which increases as the battery ages.

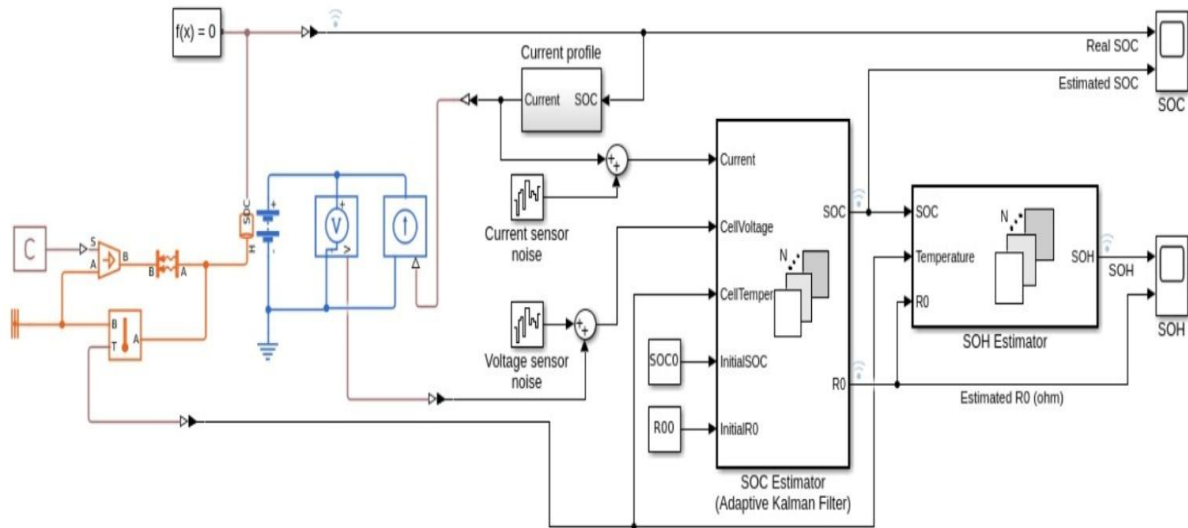


Figure 6.SOH Estimation of Batteries.

The working principle of the SOH estimator is based on monitoring battery degradation indicators over time. As the battery undergoes repeated charge–discharge cycles, its internal resistance (R_0) increases and capacity decreases, which directly affects SOH. The adaptive Kalman Filter uses this variation in R_0 along with SOC and temperature data to estimate the health condition of the battery in real time. The output of the SOH block provides a continuous estimation of battery health, which is crucial for predicting battery life, ensuring safe operation, and optimizing performance. This integrated SOC–SOH estimation model is highly useful in advanced battery management systems, as it enables accurate monitoring of both charge status and long-term degradation in electric vehicle applications.

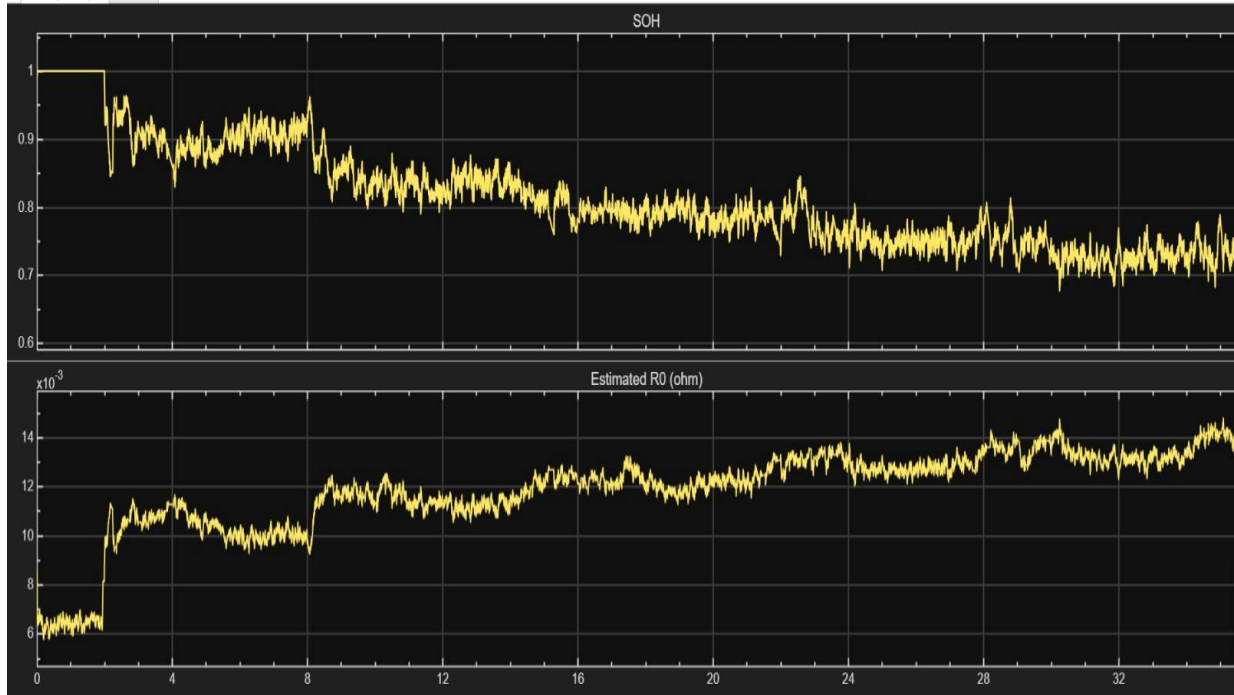


Figure 7. SOH Output.

The given output in **Figure 7** shows the **State of Health (SOH)** and the **estimated internal resistance (R0)** of the battery obtained from your adaptive Kalman Filter model in MATLAB Simulink. In the top plot, the SOH starts close to 1 (or 100%), indicating a healthy battery, and gradually decreases over time. This downward trend represents battery degradation due to repeated charge–discharge cycles. The small fluctuations observed in the curve are due to measurement noise and dynamic operating conditions, which are intentionally included in your model to reflect real-world behaviour. Overall, the decreasing trend confirms that the estimator is correctly tracking the aging process of the battery.

In the bottom plot, the estimated internal resistance (R0) shows an increasing trend over time, which is a key indicator of battery degradation. As the battery ages, its internal resistance increases, leading to higher losses and reduced efficiency. The gradual rise in R0 from a lower value to a higher value directly correlates with the decrease in SOH observed in the top plot. The small variations in the resistance curve are due to noise and transient conditions, but the overall increasing pattern validates the effectiveness of the Kalman Filter in capturing battery parameter changes. Together, these results demonstrate that your model is successfully estimating both SOH and internal resistance, making it suitable for monitoring battery health in electric vehicle applications.

D. CHARGING OF ELECTRIC VEHICLE BATTERY PACK

The Constant Current–Constant Voltage (CCCV) charging algorithm is the most widely used method for charging lithium-ion battery packs in electric vehicles. In this method, charging is performed in two stages to ensure both fast charging and battery safety. In the first stage, known as the Constant Current (CC) phase, the battery is charged with a fixed current while

the battery voltage gradually increases. This continues until the battery voltage reaches a predefined maximum limit (for example, 4.2 V per cell).

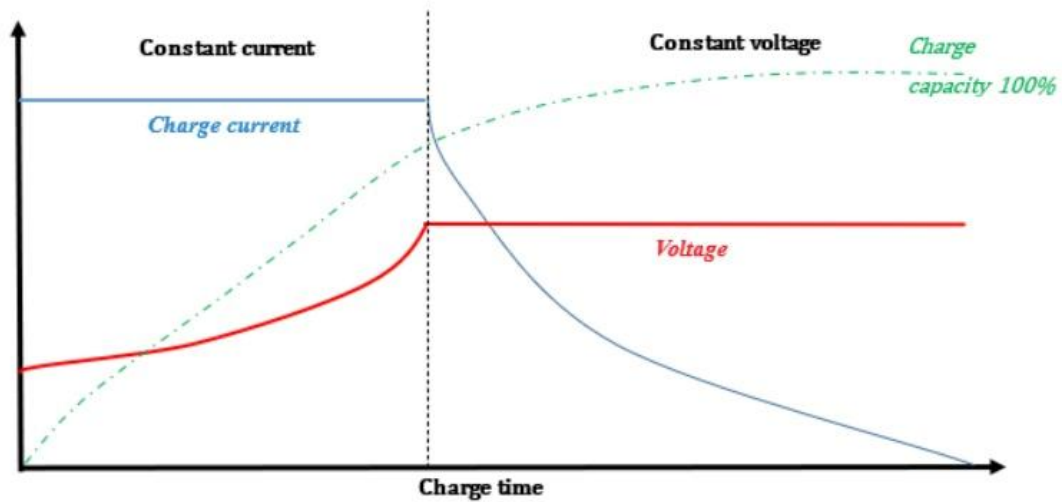


Figure 8.CCCV Model Graph.

Once this voltage threshold is reached, the charging mode switches to the second stage, called the Constant Voltage (CV) phase, where the voltage is held constant and the charging current gradually decreases. The charging process is terminated when the current falls below a small cutoff value, indicating that the battery is fully charged shown in **Figure 8**.

In MATLAB and Simulink, the CCCV charging algorithm can be implemented using control logic and basic blocks in **Figure 9**. The battery pack is connected to a controlled current source or DC-DC converter, and both voltage and current sensors are used to monitor battery conditions. During the CC phase, a constant current reference is applied using a current controller. A comparator checks when the battery voltage reaches the maximum limit, and at that point, a switch transfers control to the CV phase. In the CV phase, a voltage controller maintains the battery voltage at the set value, while the current naturally decreases. Termination logic is added using a condition that stops charging when the current drops below a specified threshold. This implementation ensures safe, efficient, and reliable charging of battery packs and is commonly used in battery management systems for electric vehicle applications.

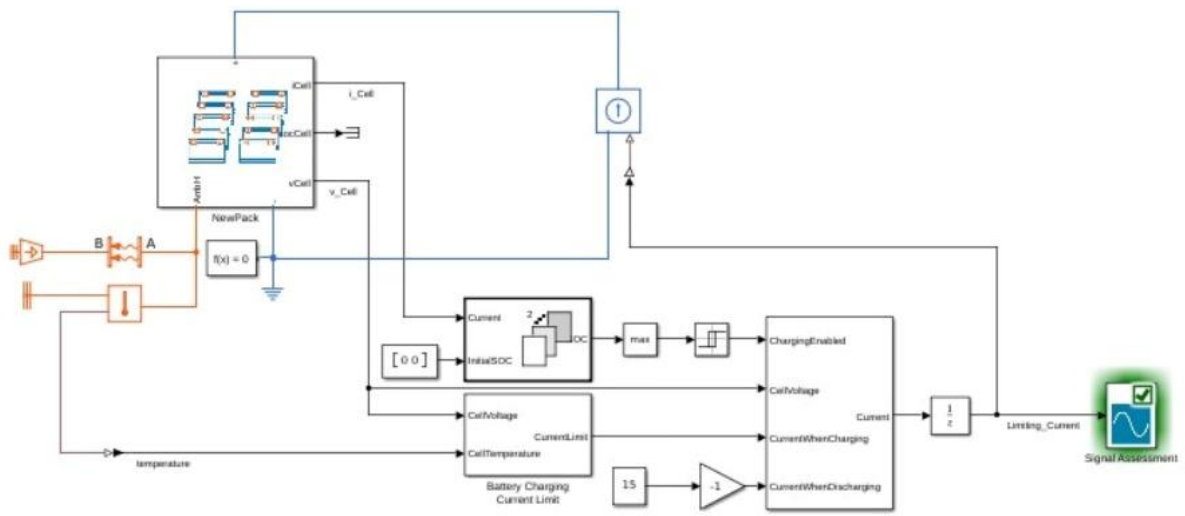


Figure 9.Charging Algorithm Using CCCV Method

The Constant Current–Constant Voltage (CCCV) charging algorithm represents how the battery voltage, current, and State of Charge (SOC) vary during the charging process in MATLAB or Simulink simulations. In the Constant Current (CC) phase, the charging current remains fixed at a predefined value, which appears as a flat (constant) line in the current waveform. During this period, the battery voltage gradually increases as the battery stores energy, and the SOC rises steadily. This phase continues until the battery voltage reaches the maximum allowable limit (e.g., 4.2 V per cell). The CC phase ensures fast charging while maintaining controlled current flow.

Once the voltage reaches its threshold, the system transitions to the Constant Voltage (CV) phase. In this phase, the voltage remains constant (appearing as a flat line in the voltage waveform), while the charging current starts to decrease gradually. This decreasing current curve indicates that the battery is approaching full charge and can no longer accept high current safely. The SOC continues to increase but at a slower rate compared to the CC phase. Finally, when the current drops below a predefined cutoff value, the charging process stops, indicating a fully charged battery. This output behaviour confirms proper implementation of the CCCV algorithm, ensuring safe, efficient, and complete charging of the battery pack.

E. CELL BALANCING

- iii. **Passive cell balancing**
- iv. **Active cell balancing**

a. PASSIVE CELL BALANCING

Passive cell balancing is a commonly used technique in battery management systems (BMS) to maintain uniform voltage and State of Charge (SOC)(**Lim et al.**) across all cells in a battery pack. In large battery packs, such as those used in electric vehicles, individual cells may develop slight differences in capacity **Figure 10**, internal resistance, or self-discharge rates over time. These variations cause some cells to charge faster and reach higher voltages than others. Passive balancing addresses this issue by dissipating excess energy from higher-voltage cells through resistors, thereby allowing lower-voltage cells to catch up. This process ensures that all cells remain within safe operating limits and improves the overall performance and lifespan of the battery pack.

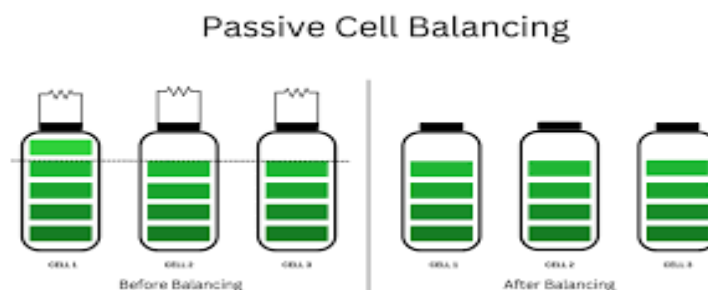


Figure 10. Passive Cell Balancing.

In practical implementation using MATLAB and Simulink, passive balancing is achieved by connecting a resistor in parallel with each cell through a controlled switch (such as a MOSFET). A control algorithm continuously monitors the voltage of each cell, and when a cell voltage exceeds a predefined threshold, the corresponding switch is turned ON, allowing current to flow through the resistor and dissipate excess energy as heat. Once the cell voltage drops to the desired level, the switch is turned OFF. This balancing process typically occurs during the charging phase and is simple, cost-effective, and easy to implement.

However, passive cell balancing has certain limitations. Since excess energy is dissipated as heat, it results in energy loss and reduced efficiency compared to active balancing methods. Additionally, the balancing process can be slow, especially for large battery packs with significant imbalance. Despite these drawbacks, passive balancing is widely used in many applications due to its simplicity, reliability, and low implementation cost, making it suitable for systems where efficiency is not the primary concern.

b. ACTIVE CELL BALANCING

Active cell balancing is a method in battery management systems where excess energy from higher-charged cells is transferred to lower-charged cells using converters or energy storage elements instead of dissipating it as heat. This approach improves efficiency, reduces energy loss, and enhances the overall performance and lifespan of the battery pack. (Lim et al.; Paidi et al. 2022)

DIFFERENT METHODS FOR ACTIVE CELL BALANCING

- Cell-to-Cell balancing
- Cell-to-Pack balancing
- Pack-to-Cell balancing
- Cell-to-Pack-to-Cell balancing
- Capacitor based balancing
- DC-to-DC converter based balancing

➤ CELL-TO-CELL BALANCING

Cell-to-Cell Balancing is an active cell balancing method in which energy is directly transferred from a higher State of Charge (SOC) cell to a lower SOC cell **Figure 11**. This transfer is typically achieved using DC-DC converters, inductors, or switching circuits that enable controlled energy flow between individual cells. By moving charge directly between cells, this method avoids energy loss in the form of heat and ensures efficient utilization of stored energy within the battery pack.

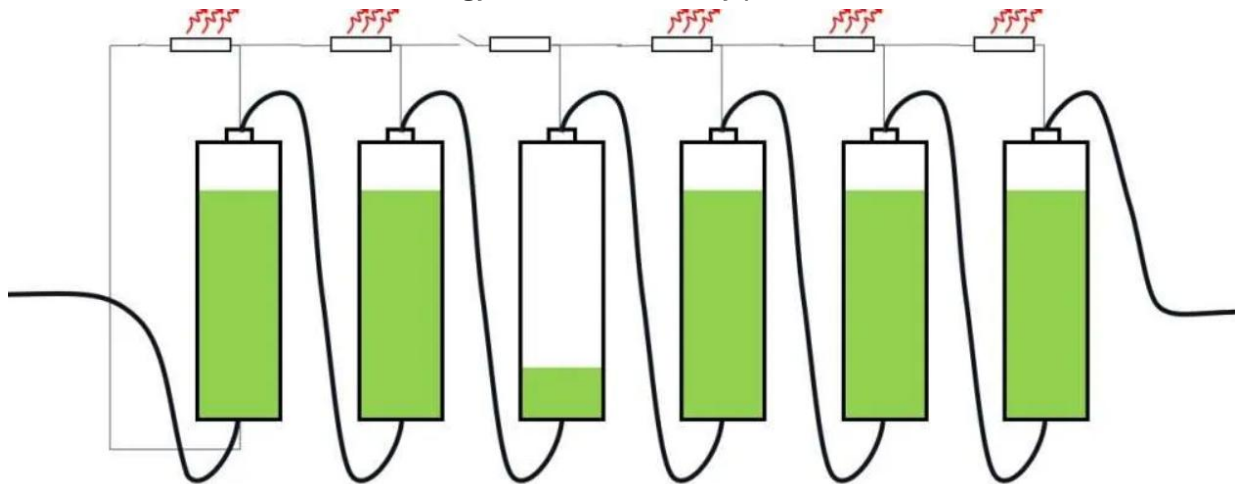


Figure 11. Cell to Cell Balancing.

This method is highly efficient and provides faster balancing compared to other techniques because it targets only the cells that require correction. It is particularly suitable for applications like electric vehicles, where maintaining uniform cell voltage is critical for

performance and safety. However, the main drawback is increased circuit complexity and control requirements, especially in large battery packs, as each cell may require dedicated switching or converter paths.

➤ **CELL-TO-PACK BALANCING**

Cell-to-Pack Balancing is an active cell balancing technique in which excess energy from a higher State of Charge (SOC) cell is transferred to the entire battery pack rather than directly to another individual cell. This is usually achieved using power electronic converters such as DC-DC converters, which extract energy from the overcharged cell and inject it into the common bus of the battery pack.

The main working principle of this method is to reduce the energy of the highest charged cell and make it available to the overall system. Once the energy is transferred to the pack, it can be naturally redistributed among other cells during normal operation. This helps in maintaining voltage uniformity across the battery pack and prevents overcharging of individual cells.

One of the key advantages of cell-to-pack balancing is its relatively simpler circuit design compared to direct cell-to-cell balancing. Since it does not require complex switching between individual cells, the control strategy becomes easier to implement. This makes it suitable for medium to large battery packs where reducing hardware complexity is important.

However, the method has slightly lower efficiency because the energy transfer is indirect. The energy taken from a high SOC cell is first sent to the pack and not directly to the weaker cell, which may lead to minor losses. Despite this drawback, cell-to-pack balancing is widely used due to its balance between simplicity, performance, and cost, making it a practical choice for many battery management system applications.

➤ **PACK-TO-CELL BALANCING**

Pack-to-Cell Balancing is an active cell balancing method in which energy is transferred from the overall battery pack to individual cells that have lower State of Charge (SOC). Instead of removing energy from higher-charged cells, this method focuses on supplying additional energy to weaker or undercharged cells. This is typically implemented using DC-DC converters that draw energy from the main pack and deliver it to specific cells.

The working principle involves continuously monitoring the voltage or SOC of each cell in the battery pack. When a cell is identified as having a lower SOC compared to others, the control system activates the converter to direct energy from the pack to that particular cell. This helps in increasing the charge level of weaker cells and maintaining uniformity across the entire battery pack.

One major advantage of pack-to-cell balancing is that it effectively improves the performance of weaker cells without disturbing the energy levels of stronger cells. It is particularly useful in situations where certain cells degrade faster and require additional support to match the overall pack performance. This method also helps in extending the usable capacity of the battery pack.

However, the method involves more complex control and power electronic circuits compared to simpler balancing techniques. Since energy must be selectively directed to individual cells, the system requires precise switching and monitoring mechanisms. Despite this complexity, pack-to-cell balancing is valuable in advanced battery management systems, especially in electric vehicles, where maintaining uniform cell performance is critical for safety and efficiency.

➤ **CELL-TO-PACK-TO-CELL BALANCING**

Cell-to-Pack-to-Cell Balancing is an advanced active cell balancing method that combines both cell-to-pack and pack-to-cell energy transfer techniques. In this approach, excess energy from a higher State of Charge (SOC) cell is first transferred to the battery pack, and then redistributed from the pack to a lower SOC cell. This indirect energy transfer allows balancing between non-adjacent cells without requiring direct connections between every pair of cells.

The working principle involves two stages of energy flow. In the first stage, energy is extracted from the overcharged cell and sent to a common energy bus or the battery pack using a DC-DC converter. In the second stage, this stored energy is directed from the pack to the undercharged cell using another controlled converter. This two-step transfer process ensures that energy can be efficiently shared across the entire battery pack.

One of the major advantages of this method is its flexibility. It allows balancing between any cells in the pack, regardless of their physical location, which is especially useful in large battery systems like electric vehicles. Additionally, it avoids direct cell-to-cell connections, reducing wiring complexity compared to pure cell-to-cell balancing methods.

However, the main drawback of this method is increased system complexity and cost. Since it requires multiple converters and more advanced control strategies, the design and implementation become more complicated. There may also be additional energy losses due to the two-stage conversion process. Despite these challenges, cell-to-pack-to-cell balancing is considered an effective and scalable solution for modern battery management systems.

➤ **SWITCHED CAPACITOR BASED ACTIVE CELL BALANCING**

The battery section is created using two individual battery cell blocks connected in series. Each cell should have voltage measurement blocks connected across it so that the individual cell voltages can be monitored. If SOC estimation is required, SOC blocks or battery models with built-in SOC output can be used. These signals are later used for comparison to decide which cell needs balancing.

Next, the energy transfer element, which is the capacitor, is added between the switching network. A capacitor block is placed such that it can be connected across either of the two cells through switches. The capacitor value is selected based on the required balancing speed (typically in the range of microfarads to millifarads for simulation).

After that, the switching circuit is designed using four MOSFET switches (or ideal switches in Simulink), labeled S1, S2, S3, and S4. These switches are arranged in such a way that:

- One pair (S1 and S4) connects the capacitor across Cell 1
- Another pair (S2 and S3) connects the capacitor across Cell 2

The switches must be connected carefully so that the capacitor can alternately charge from one cell and discharge into the other without creating a short circuit path between the cells.

Then, the switching control signals are generated using two pulse generators. One pulse generator controls S1 and S4 (charging phase), and the other controls S2 and S3 (discharging phase). These pulses must be complementary, meaning when one is ON, the other must be OFF. A small dead time is usually added between them to prevent both switch pairs from turning ON at the same time.

To make the system intelligent, a comparison logic is added using voltage or SOC signals. A comparator or “max” block is used to determine which cell has higher voltage. Based on this, the control logic can enable or disable balancing or even reverse the switching sequence if needed. In simple models, fixed pulses are used, while in advanced models, logic-based control is implemented.

A current sensor block is inserted in the capacitor path to measure the balancing current. This helps in analyzing how much current is flowing during charge and discharge cycles. A voltage measurement block can also be connected across the capacitor to observe its charging and discharging behavior.

To simulate realistic battery operation, a charging control block such as CC-CV (constant current–constant voltage) can be included. This block controls the charging current and voltage limits of the battery pack. A relay or switch can also be used to toggle between charging and discharging modes.

Finally, scopes are added to visualize important outputs such as:

- Cell voltages
- SOC of each cell
- Capacitor voltage
- Balancing current

When the simulation runs, the pulse generators continuously switch the capacitor between the two cells. The capacitor charges from the higher voltage cell and discharges into the lower voltage cell repeatedly. As a result, the voltage difference gradually reduces, demonstrating the balancing action.

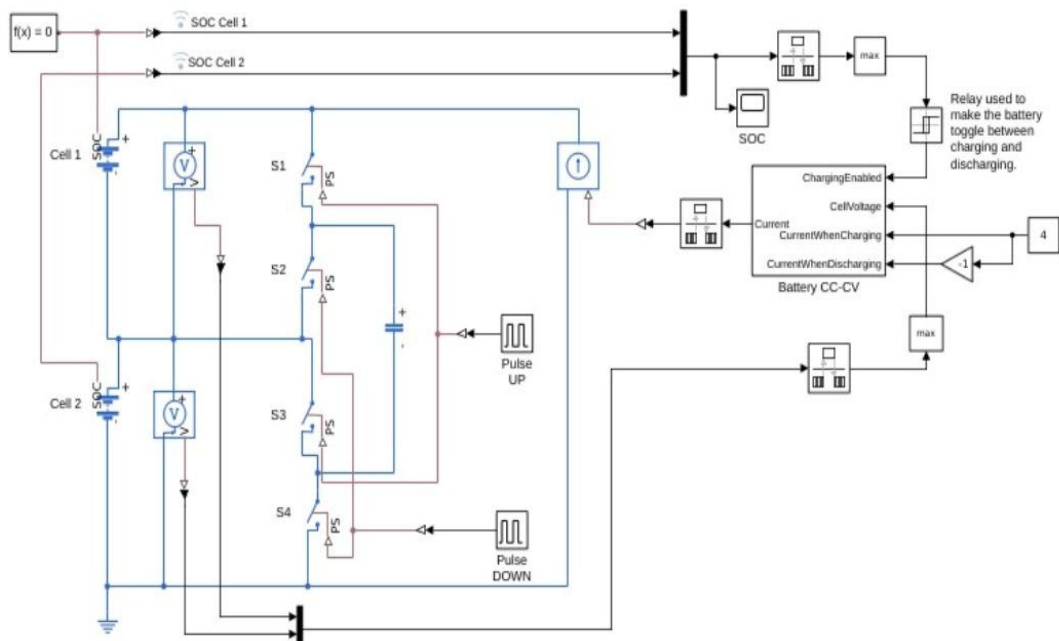


Figure 12, Switched Capacitor Based Active Cell Balancing.

The switched capacitor based active cell balancing circuit works by transferring energy between adjacent battery cells using a capacitor and controlled switching. In the given system, two cells are continuously monitored for their voltage or state of charge (SOC). When a difference is detected, the circuit automatically starts the balancing process by identifying which cell has higher energy.

Initially, the capacitor is connected across the higher voltage cell through a specific combination of switches (S1 and S4). In this condition, the capacitor charges up to the voltage

level of that cell, storing energy temporarily. Once the capacitor is fully or sufficiently charged, the switching state changes.

In the next step, the capacitor is disconnected from the higher voltage cell and connected across the lower voltage cell using another set of switches (S2 and S3). Since the capacitor voltage is now higher than the lower voltage cell, energy flows from the capacitor into that cell, effectively increasing its charge.

This process of charging the capacitor from the higher energy cell and discharging it into the lower energy cell is repeated continuously using pulse signals generated by the UP and DOWN controllers. These pulses ensure proper timing and prevent overlapping of switches, which could otherwise cause short circuits.

As the switching continues, energy is gradually redistributed between the two cells. The higher voltage cell loses some charge, while the lower voltage cell gains charge. Over time, the voltage and SOC difference between the cells reduces, and both cells reach a balanced state.

During this operation, the current flowing through the circuit appears in pulses due to the switching action, and the capacitor voltage alternates depending on which cell it is connected to. The overall system operates efficiently because energy is transferred rather than dissipated as heat.

Thus, the switched capacitor method achieves cell balancing by repeatedly moving charge between cells through controlled switching, ensuring improved battery performance and longer lifespan.



Figure 13. Output of Switched Capacitor-Based Cell Balancing.

The graph in **Figure 13**, shows the SOC variation of two cells (soc1 and soc2) during the switched capacitor balancing process, and it clearly demonstrates how the balancing circuit is working.

Initially, there is a noticeable imbalance between the two cells. One cell starts at a lower SOC (around 0.70), while the other is at a higher SOC (around 0.75). This difference indicates that one cell has more stored energy than the other, which triggers the balancing operation.

As the simulation begins, both SOC curves increase steadily. This indicates that the battery pack is in the charging phase, where external charging is applied (through the CC-CV block in your model). Even during charging, the switched capacitor circuit is active, transferring energy from the relatively higher SOC cell to the lower SOC cell. Because of this, although both SOC increase, the gap between them does not widen significantly.

Around the midpoint of the graph (near SOC $\approx$ 0.88–0.90), both cells reach their peak values. At this stage, the SOC difference between the two cells has reduced compared to the initial condition. This shows that the balancing circuit is effectively redistributing energy while charging is happening.

After this point, both SOC values start decreasing, indicating the discharging phase of the battery pack. During discharging, the balancing circuit continues to operate. The higher SOC cell supplies slightly more energy not only to the load but also indirectly to the lower SOC cell through the capacitor switching action. Because of this, both SOC curves decrease in a nearly parallel manner.

An important observation is that throughout the entire simulation, the SOC curves of both cells move closer together compared to their initial difference. The gap between soc1 and soc2 is reduced, which confirms that energy is being successfully transferred from the higher energy cell to the lower energy cell.

The smooth and nearly parallel nature of the curves also indicates stable operation of the switched capacitor circuit. There are no sudden jumps or oscillations, which means the switching pulses are properly synchronized and the system is operating efficiently.

➤ DC TO DC CONVERTER-BASED CELL BALANCING

In dc-to-dc converter based active cell balancing. this balancing is one of the most effective methods in recent times for the active cell balancing this method is widely using in the high-end models like tesla, BMW etc., in this method we can control every cell power by using the electronic components like PWM, bidirectional dc-dc converter, PI controller are the most important components in this method.

Active cell balancing using a dc–dc converter is an efficient technique used in battery management systems to maintain equal voltage and state of charge (soc) among all the cells in a battery pack. In large battery systems, such as those used in electric vehicles or energy storage applications, individual cells tend to develop differences in voltage due to manufacturing variations, temperature effects, and aging. If these imbalances are not corrected, they can reduce the overall performance, usable capacity, and lifespan of the

battery pack. The dc–dc converter-based active balancing method addresses this issue by redistributing energy from higher-charged cells to lower-charged cells instead of wasting it as heat, which makes it significantly more efficient than passive balancing methods.

The working principle of this method is based on controlled energy transfer using power electronic components. The battery management system continuously monitors the voltage of each cell and identifies cells with higher and lower charge levels. A bidirectional dc–dc converter is then used to transfer energy between these cells. When a high-voltage cell is selected, the converter operates in buck mode to step down the voltage and extract energy. This energy is temporarily stored in an inductor or capacitor and then delivered to a lower-voltage cell by operating the converter in boost mode. The switching of the converter is controlled using pulse width modulation (pwm), ensuring smooth and efficient energy transfer without significant losses.

The circuit used in this method typically consists of inductors, capacitors, mosfet switches, and a gate driver circuit. The inductor plays a crucial role in storing and transferring energy, while capacitors help in reducing voltage ripple and stabilizing the output. Mosfets act as high-speed switches controlled by the gate driver, which receives signals from the bms controller. The design of these components is important for proper operation. For example, a switching frequency of around 50 khz is commonly used to balance efficiency and switching losses. The inductor value is generally selected in the range of 47 μ h to 220 μ h, with 100 μ h being a typical choice. Similarly, capacitors of around 220 μ f are used to ensure stable voltage during energy transfer.

In practical implementation, the balancing current is usually maintained between 0.5 a and 2 a, depending on the battery size and application requirements. The mosfets used in the circuit must have sufficient voltage and current ratings, typically above 30 v and 10 a, to ensure safe operation. The balancing process is initiated when the voltage difference between cells exceeds a certain threshold, such as 20 mv. Once activated, the converter transfers energy until the voltage difference reduces to an acceptable level. This controlled operation ensures that all cells reach a balanced state without overcharging or deep discharging any individual cell.

One of the major advantages of the dc–dc converter-based active balancing method is its high efficiency, which can reach up to 90% or more. Since energy is redistributed rather than dissipated, the system generates less heat and improves overall battery performance. This method also enables faster balancing compared to passive techniques and is highly suitable for large battery packs with hundreds of cells. However, the system is more complex due to the need for multiple switching components and control algorithms. It also increases the cost and design complexity, especially in large-scale implementations where multiple converters or modular designs may be required.

Active cell balancing using a dc–dc converter is a highly effective and modern approach for maintaining uniform cell voltage in battery packs. By enabling efficient energy transfer between cells, it enhances battery life, improves safety, and maximizes usable capacity. Although the design is more complex than traditional methods, its advantages make it an essential technique in advanced battery systems, particularly in electric vehicles and renewable energy storage applications.

We designed active cell balancing for the battery pack build by battery builder app in MATLAB . The battery is like practical battery pack electric vehicle insuch that we can implement total active cell balancing control for the entire pack.

- **BATTERY PACK REALISTIC BEHAVIOR FOR BETTER ACTIVE CELL BALANCING**

A battery pack is a structured combination of many individual battery cells connected in series and parallel to deliver the required voltage, current, and capacity for practical applications. Unlike a single battery cell, which typically provides only a small amount of voltage (around 3.7 V for lithium-ion) and limited capacity, a battery pack is designed to meet real-world power demands such as those in electric vehicles, renewable energy storage systems, and portable electronics. The pack is not just a collection of cells; it also includes important components like a battery management system (BMS), thermal management system, sensors, and protective circuitry. These additional elements make the battery pack behave like a complete energy system rather than a simple energy source.

Table 3.Simulink Library Path and Configurations

Block Name	Simulink Library Path	Configuration
Li-Ion Cell Model	Simscape > Electrical > Batteries > Generic Battery (Simscape)	Set: Vnom=3.7V, Cap=50Ah, internal resistance=0.003Ω, RC pairs=2 for dynamics
Cell Array (30S×30P)	Simscape Electrical: use Series/Parallel connection	Copy cell block ×900, arrange in 30 series groups each with 30 parallel branches
Cell Voltage Sensor	Simscape > Foundation Library > Electrical > Electrical Sensors > Voltage Sensor	Place one across each cell — 900 total. Bus all outputs into Vcell [1...900] bus
Cell Temperature (Thermal)	Simscape > Thermal > Thermal Mass + Heat Source	Add thermal domain per cell: R_thermal=2°C/W, C_thermal=50J/°C
Pack Current Sensor	Simscape > Foundation Library > Electrical > Current Sensor	Place in series with pack output — single sensor for I_pack

Temperature Sensor	Simscape > Foundation Library > Thermal > Temperature Sensor	One per cell, output Tcell [1...900]
Bus Creator (Voltage)	Simulink > Signal Routing > Bus Creator	900 inputs → single bus Vcell [1...900]
Bus Creator (Temp)	Simulink > Signal Routing > Bus Creator	900 inputs → single bus Tcell [1...900]

Battery packs are considered more realistic than normal individual batteries because they reflect how energy storage systems are actually used in real applications. In practical scenarios, a single cell cannot supply sufficient power or energy for long durations, so multiple cells are combined to achieve the desired performance. However, when many cells are connected together, variations naturally occur due to manufacturing differences, temperature gradients, and aging effects. These variations cause imbalance in voltage and state of charge among cells, which is a real-world problem that does not appear when studying a single battery. Therefore, battery packs provide a more accurate and practical representation of real operating conditions compared to individual batteries.

This realistic behaviour makes battery packs highly relevant for studying and implementing active cell balancing techniques. In a battery pack, imbalance between cells can lead to reduced overall capacity, overheating, and even safety risks such as overcharging or deep discharging of certain cells. Active cell balancing becomes essential in such systems because it redistributes energy among cells to maintain uniform voltage levels. Compared to simple or normal batteries, battery packs benefit much more from active balancing since the energy differences between cells are significant and continuous during operation. Without balancing, the weakest cell limits the performance of the entire pack.

Table 4. Cell Parameters

Parameter	Value / Expression
Nominal Voltage (Vnom)	3.7 V
Maximum Voltage (Vmax)	4.2 V
Minimum Voltage (Vmin)	2.7 V
Rated Capacity	50 Ah
Series Resistance (R0)	0.003 Ω
RC Pair 1: R1, C1	0.005 Ω , 2000 F
RC Pair 2: R2, C2	0.010 Ω , 5000 F
Initial SOC	Vary per cell: 0.5 to 0.9 (use rand for spread)
Self-Discharge	0.1% / month = 3.86e-8 per second

Furthermore, battery packs allow active cell balancing methods, such as DC–DC converter-based techniques, to be effectively utilized. Since there are multiple cells, energy can be transferred from higher-energy cells to lower-energy ones, improving overall efficiency. In contrast, a single battery or small system does not provide enough scope for such energy redistribution. The bus connection ports will give the all battery pack information from monitor. and these buses are called can buses . this buses plays most important role in battery management. The battery management system also having main thing is cell balancing for that also we need can bus .

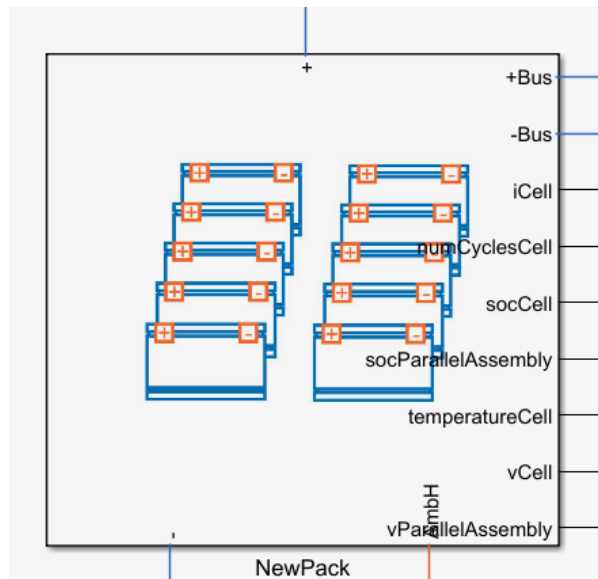


Figure 14. Table 5. Battery Pack Build Using Battery Builder App.

Active balancing in battery packs helps in increasing usable capacity, improving efficiency, reducing heat losses, and extending the overall lifespan of the system. This makes battery packs not only more realistic but also more suitable and beneficial for advanced balancing techniques compared to normal individual batteries.

• SOC ESTIMATION USING KALMAN FILTER FOR BATTERY PACK

State of Charge (SOC) estimation using a Kalman Filter in active cell balancing, especially in a DC–DC converter-based method, arises from the limitations of simple voltage-based balancing techniques. In a battery pack, the voltage of a cell does not always accurately represent its actual charge level due to effects like internal resistance, temperature variation, and load conditions. As a result, relying only on voltage can lead to incorrect identification of which cells need balancing. This is where SOC estimation becomes essential, as it provides a more accurate representation of the available charge in each cell.

The Kalman Filter is a powerful mathematical algorithm used to estimate the internal states of a system when direct measurement is not possible or is noisy. In battery

systems, it combines measurements such as current and voltage with a mathematical model of the battery to estimate SOC in real time. It continuously corrects its estimation by comparing predicted values with actual measurements, making it highly accurate even under dynamic operating conditions. This makes it more reliable than traditional methods like Coulomb counting, which accumulates errors over time.

Table 5. MATLAB Library Blocks

Block Name	Library / Type	Settings
MATLAB Function (EKF Core)	Simulink > User-Defined Functions > MATLAB Function	Implement state equation: $x=[SOC, V_1, V_2]$, predict update steps. See code template below
Unit Delay (State Memory)	Simulink > Discrete > Unit Delay	Stores previous state vector $[SOC, V_{RC1}, V_{RC2}]$. IC = $[0.7; 0; 0]$ per cell
OCV Lookup Table	Simulink > Lookup Tables > 1-D Lookup Table	Map SOC (0-1) to OCV (2.7-4.2V). Use measured OCV-SOC curve data
For Each Subsystem	Simulink > Ports & Subsystems > For Each Subsystem	Wrap EKF to run independently on each of 900 cells
Mux (EKF inputs)	Simulink > Signal Routing > Mux	Combine Vcell, Icell, Tcell per iteration

In the context of DC–DC converter-based active cell balancing, accurate SOC estimation is critical for proper decision-making. The balancing system must identify which cell has excess energy and which one requires energy. If SOC is not estimated correctly, the system may transfer energy in the wrong direction, leading to inefficiency or even worsening the imbalance. By using a Kalman Filter, the BMS can precisely detect small differences in SOC between cells and initiate balancing only when necessary, improving overall system performance.

Another important reason for using a Kalman Filter is its ability to handle noise and uncertainties in measurements. In real battery packs, sensor readings are often affected by electrical noise and environmental conditions. The Kalman Filter minimizes these errors and provides a smooth and reliable SOC estimate. This ensures stable control of the DC–DC converter, as switching decisions depend heavily on accurate input data. Without such filtering, the converter may operate inefficiently due to fluctuating or incorrect signals.

- **PI CONTROLLER AND PWM**

In a DC–DC converter-based active cell balancing system, the PI (Proportional–Integral) controller and PWM (Pulse Width Modulation) work together to regulate the amount of energy transferred between cells. The PI controller continuously monitors the error between the desired balancing condition (reference value, usually zero SOC difference or target current) and the actual condition (measured SOC difference or balancing current).

Table 6.Controller Blocks

Block Name	Library	Settings
PID Controller Block	Simulink > Continuous > PID Controller	Set: $P=K_p=2.0$, $I=K_i=0.5$, $D=0$. Form: Parallel. Init condition = 0
Gain K_p	Simulink > Math Operations > Gain	Value = 2.0 (tunable). Proportional path
Integrator K_i/s	Simulink > Continuous > Integrator	Gain $K_i=0.5$ at input. IC=0. Add anti-windup: limit state to ± 10
Sum ($K_p + K_i/s$)	Simulink > Math Operations > Sum	Add proportional and integral outputs. Output = I_{ref}
Saturation (I_{ref})	Simulink > Discontinuities > Saturation	Upper = +10A, Lower = -10A. Limits balancing current
Sort Block	Simulink > Math Operations > Sort OR MATLAB Function	Sorts SOC array to determine which cells are most deviant. Feeds sorted indices to PI input

Based on this error, the proportional part provides an immediate response, while the integral part accumulates past errors to eliminate steady-state deviation. As a result, the PI controller generates a control signal that determines how much energy should be transferred from the source cell to the target cell.

This control signal is then fed to the PWM generator, which converts it into a switching signal with a specific duty cycle. The duty cycle represents the ON and OFF time of the switches in the DC–DC converter. A higher duty cycle means more energy transfer, while a lower duty cycle reduces the transfer. The PWM signal is applied to the gate of the source switch, while its complement is applied to the target switch, ensuring proper operation of the converter. By continuously adjusting the duty cycle based on PI controller output, the system achieves smooth, efficient, and controlled balancing.

Overall, the PI controller ensures accurate control of balancing current or SOC difference, while PWM provides the practical switching mechanism to implement that control in hardware. Together, they enable stable, efficient, and precise energy transfer between cells, improving battery performance and lifespan.

- **CELL SELECTION LOGIC**

Cell selection logic is used in a battery management system (BMS) to decide which cell (or group of cells) should transfer energy and which cell should receive energy during active cell balancing using a DC–DC converter. This logic is very important because proper selection of source and target cells ensures efficient balancing, improves battery life, and maintains safety. The function takes four inputs: state of charge (SOC), state of health (SOH), voltage, and temperature of each cell group, and produces outputs indicating the source cell, target cell, SOC difference, and system status flags.

The function begins by defining system parameters such as the number of cells ($N = 10$), maximum and minimum voltage limits (4.2 V and 2.5 V), temperature limits (45°C and -10°C), and a SOC imbalance threshold of 0.05 (i.e., 5%). These parameters act as safety and operational boundaries. It also introduces a tuning factor called alpha, which is used to incorporate the effect of SOH into SOC calculation. This is important because cells with lower health should be treated differently even if their SOC appears similar.

Next, the function initializes all output variables. The source index (`src_idx`) and target index (`tgt_idx`) are initially set to 1, the SOC difference (`delta_SOC`) is set to zero, and both `enable_flag` and `fault_flag` are set to false. This ensures a safe default condition before any computation is performed. After initialization, a safety check is performed to verify whether the SOC array length matches the expected number of cells. If there is a mismatch, the function immediately sets the `fault_flag` to true and exits, preventing incorrect operation.

The core concept of this function lies in the calculation of SOH-weighted effective SOC. Instead of using SOC directly, the function modifies it using the formula: $\text{effective SOC} = \text{SOC} + \alpha \times (1 - \text{SOH})$. This means that cells with poor health (low SOH) are given slightly higher priority in balancing decisions. This approach helps in maintaining long-term battery performance by preventing weaker cells from becoming more stressed.

After computing the effective SOC, the function identifies the cell with the maximum value and the cell with the minimum value. The cell with the highest effective SOC is selected as the source cell, meaning it has excess energy to give. The cell with the lowest effective SOC is selected as the target cell, meaning it needs energy. The difference between these two values is calculated as `delta_SOC`, which indicates how much imbalance exists in the system.

Table 7. Cell Parameters and Symbols

Parameter	Symbol/Variable	Value / Description
Number of cells	N	10 (series groups)
Maximum voltage	V_MAX	4.2 V
Minimum voltage	V_MIN	2.5 V
Maximum temperature	T_MAX	45 °C
Minimum temperature	T_MIN	-10 °C
SOC threshold	SOC_THRESH	0.05 (5% imbalance)
SOH weighting factor	alpha	0.1 (tuning parameter)
Source index	src_idx	Cell with max effective SOC
Target index	tgt_idx	Cell with min effective SOC
SOC difference	delta_SOC	
Enable flag	enable_flag	TRUE if balancing required
Fault flag	fault_flag	TRUE if safety limits violated
Effective SOC	eff	$SOC + \alpha(1 - SOH)$

The function then applies protection logic to ensure safe operation. It checks whether any cell voltage exceeds the maximum limit or drops below the minimum limit, and whether any temperature is outside the safe range. If any of these conditions are violated, the system sets the `fault_flag` to true. This prevents the balancing process from operating under unsafe conditions, which could otherwise damage the battery or create hazardous situations.

Following the protection check, the function determines whether balancing should be enabled. The `enable_flag` is set to true only if three conditions are satisfied: there is no fault, the SOC difference exceeds the defined threshold (5%), and the source and target cells are not the same. This ensures that balancing occurs only when it is necessary and safe, avoiding unnecessary switching and energy loss.

Finally, a fault-safe override mechanism is implemented. If a fault is detected, the function resets the source and target indices to 1 and sets `delta_SOC` to zero. This effectively disables balancing and brings the system to a safe state. This step is crucial in real-world applications, as it ensures that no energy transfer occurs during abnormal conditions.

In summary, this cell selection logic intelligently chooses which cells to balance based on SOC, SOH, voltage, and temperature. It ensures accurate, efficient, and safe operation of

the DC–DC converter-based active cell balancing system by combining performance optimization with strict safety check.

- **SWITCHING LOGIC**

The given switching logic function is responsible for generating the gate signals for the DC–DC converter switches in an active cell balancing system. While the cell selection logic decides *which cells* to balance, this switching logic determines how the power switches (MOSFETs) should operate to transfer energy from the source cell to the target cell. It converts high-level decisions (source index, target index, enable signal) into actual switching signals required by the hardware or Simulink model.

The function takes five inputs: source cell index (src_idx), target cell index (tgt_idx), enable flag, PWM signal, and fault flag. Based on these inputs, it produces three outputs: G_hi (high-side gate signals), G_lo (low-side gate signals), and sw_status (switching status indicator). The system is designed for 10 cell groups (N = 10), and for each group, separate gate signals are generated. Initially, all gate signals are set to false (OFF), and the system status is set to idle (0), ensuring safe startup conditions.

Table 8. Switch parameters and Symbols

Parameter	Input Values (Example)	Output / Result
Source Index	4	G_hi(4) = PWM = 1
Target Index	6	G_lo(6) = ~PWM = 0
Enable Flag	TRUE	Switching allowed
PWM Signal	1 (ON state)	Controls energy transfer
Fault Flag	FALSE	No fault condition
G_hi (10×1)	[0 0 0 1 0 0 0 0 0 0]	Only source active
G_lo (10×1)	[0 0 0 0 0 0 0 0 0 0] → (for PWM=1)	Target OFF (complement logic)
Switching Status	—	1 (Active Balancing)

The first step in the logic is the safety check, which plays a critical role in protecting the battery and converter. If a fault is detected, or if balancing is not enabled, or if the source and target cells are the same, the function immediately stops further execution. In this case, all switches remain OFF, and no energy transfer occurs. If the fault_flag is true, the switching status is set to 2, indicating a fault condition. This prevents any unsafe switching operations and ensures the system operates only under valid conditions.

Once the system passes the safety check, the function implements a one-hot decoding strategy. This means that at any given time, only one source cell and one target cell are actively involved in switching. For the selected source cell (src_idx), the corresponding high-side gate signal (G_hi) is driven by the PWM signal. This means the source cell is connected to the converter in a controlled switching manner, allowing energy to be extracted. For the target cell (tgt_idx), the low-side gate signal (G_lo) is driven by the complement of the PWM signal ($\sim$ PWM_signal). This complementary switching ensures proper current flow through the converter and avoids short circuits.

The use of PWM (Pulse Width Modulation) is important because it controls the amount of energy transferred between cells. By adjusting the duty cycle of the PWM signal, the system can regulate the balancing current. A higher duty cycle results in more energy transfer, while a lower duty cycle reduces it. The complementary switching between G_hi and G_lo ensures efficient operation of the DC–DC converter, typically forming a buck-boost energy transfer mechanism between the selected cells.

Table 9.Switching Input Values and Output Results

Parameter	Input Values (Example)	Output / Result
Source Index	4	G_hi(4) = 0
Target Index	6	G_lo(6) = 1
Enable Flag	TRUE	Switching allowed
PWM Signal	0 (OFF state)	Complement switching active
Fault Flag	FALSE	No fault
G_hi (10×1)	[0 0 0 0 0 0 0 0 0 0]	Source OFF
G_lo (10×1)	[0 0 0 0 0 1 0 0 0 0]	Target ON
Switching Status	—	1 (Active Balancing)

The function updates the switching status. If the system is operating normally and balancing is active, the status is set to 1, indicating active balancing. If no balancing is happening, it remains 0 (idle), and if a fault occurs, it is set to 2. This status output is useful for monitoring and debugging the system in simulations or real-time implementations.

Table 10. Switching Parameters Inputs and Output Values

Input Conditions	Selected Cells	System Output
SOC imbalance: Max = Cell 4 (0.92), Min = Cell 6 (0.76)	Source = 4, Target = 6	$\Delta\text{SOC} = 0.16 (>0.05)$, Enable = TRUE
Voltage & Temp within limits	—	Fault = FALSE
PWM = 1	—	G_hi(4)=1, G_lo(6)=0
PWM = 0	—	G_hi(4)=0, G_lo(6)=1
Fault condition (e.g., $V > 4.2\text{V}$)	—	All switches OFF, Status = Fault

In summary, the switching logic acts as the execution layer of the active balancing system. It translates the decisions made by the cell selection logic into precise switching signals required to operate the DC–DC converter. By ensuring safe operation, controlled energy transfer using PWM, and selective activation of source and target cells, this logic plays a crucial role in achieving efficient and reliable active cell balancing.

- **GATE DRIVER**

A gate driver is a power amplifier that accepts a low-power input from a controller IC and produces a high-current drive input for the gate of a high-power transistor such as an IGBT or power MOSFET. Gate drivers can be provided either on-chip or as a discrete module. In essence, a gate driver consists of a level shifter in combination with an amplifier. A gate driver IC serves as the interface between control signals (digital or analog controllers) and power switches (IGBTs, MOSFETs, SiC MOSFETs, and GaN HEMTs). An integrated gate-driver solution reduces design complexity, development time, bill of materials (BOM), and board space while improving reliability over discretely-implemented gate-drive solutions.

The given gate_driver function acts as an interface between the switching logic and the power MOSFETs in the DC–DC converter used for active cell balancing. While the switching logic generates digital signals (0 or 1), MOSFETs require proper analog gate voltages to turn ON and OFF. This function converts those logical signals into actual gate drive voltages and ensures safe operation of the converter.

The function takes two inputs: G_hi and G_lo, which are Boolean signals representing the switching commands for the high-side and low-side MOSFETs. These signals are generated from the PWM-based switching logic. The gate driver then converts these logical values into voltage levels using predefined parameters: $V_{\text{ON}} = 15\text{ V}$ and $V_{\text{OFF}} = 0\text{ V}$. When

the input signal is 1 (true), the corresponding MOSFET gate receives 15 V, which fully turns it ON. When the input is 0 (false), the gate voltage is set to 0 V, turning the MOSFET OFF. This conversion is important because MOSFETs require sufficient gate voltage (typically 10–15 V) to operate efficiently with low resistance.

Table 11. Gate Drivers

Block Name	Library	Settings
NOT Gate (complement)	Simulink > Logic & Bit Ops > Logical Operator	Inverts PWM to get complementary gate signal for low-side switch
Transport Delay (deadtime)	Simulink > Continuous > Transport Delay	Delay = 100ns (1e-7 s). Apply to BOTH high and low side to create deadtime and prevent shoot-through
AND Gate (enable)	Simulink > Logic & Bit Ops > Logical Operator	ANDs gate signal with safety_enable. Shuts off all gates on fault
Gain (15V level)	Simulink > Math Operations > Gain	Gain = 15. Converts 0/1 PWM to 0V/15V gate voltage

An important feature of this gate driver is dead-time or shoot-through protection. In power converters, if both high-side and low-side MOSFETs turn ON at the same time, it creates a direct short circuit across the supply, known as shoot-through, which can severely damage the circuit. To prevent this, the code checks if both `G_hi` and `G_lo` are true simultaneously using the condition `both_on = G_hi & G_lo`. If such a condition occurs, the function forces both gate voltages to zero, effectively turning OFF both switches and avoiding any short circuit.

This safety mechanism ensures reliable operation of the DC–DC converter during fast switching conditions. In practical systems, a small delay (dead-time) is usually introduced between turning OFF one switch and turning ON the other, but in this simplified implementation, the logic directly disables both switches in case of conflict. This makes the system robust and suitable for simulation as well as real-time control.

Overall, the gate driver plays a crucial role in translating control signals into actual switching actions. It ensures that the MOSFETs operate with proper voltage levels and protects the system from dangerous conditions like shoot-through. In the context of active cell balancing, this helps achieve safe, efficient, and controlled energy transfer between cells using the DC–DC converter.

- **INDUCTOR BASED DC-DC CONVERTER**

An inductor-based DC–DC converter is a power electronic circuit(Lee et al.) used to convert one DC voltage level to another using energy storage elements like inductors and capacitors. In active cell balancing systems, this type of converter plays a crucial role in transferring energy efficiently from one battery cell to another. The key advantage of using an inductor is its ability to store energy in the form of a magnetic field when current flows through it and release that energy when required. This property makes it ideal for controlled energy transfer between cells without significant losses.

The design of an inductor-based DC–DC converter mainly involves selecting appropriate values for the inductor, switching frequency, capacitor, and power switches (MOSFETs). The inductor value is chosen based on the desired current ripple and switching frequency, typically ranging from 47 μH to 220 μH in battery applications. A switching frequency of around 50 kHz is commonly used to balance efficiency and switching losses. Capacitors, usually in the range of 100 μF to 470 μF , are used to smooth the output voltage and reduce ripple. MOSFETs are selected with suitable voltage and current ratings to handle the power levels safely. Proper design ensures stable operation, minimal losses, and efficient energy transfer.

Table 12.Power Electronics

Block Name	Library	Settings
MOSFET Q1 (High-side)	Simscape > Electrical > Semiconductors > MOSFET	Ron=0.002 Ω , Roff=1e6 Ω , Vth=0. Gate driven by PWM_high = duty
MOSFET Q2 (Low-side)	Simscape > Electrical > Semiconductors > MOSFET	Ron=0.002 Ω . Gate driven by PWM_low = 1-duty (complementary with deadtime)
Inductor L	Simscape > Foundation > Electrical > Inductor	L = 100 μH , R_series=0.01 Ω . Stores and transfers energy between cells
Diode D1, D2 (Body diodes)	Simscape > Electrical > Semiconductors > Diode	Vf=0.7V, Ron=0.01 Ω . Connected antiparallel to each MOSFET
Capacitor C_in	Simscape > Foundation > Electrical > Capacitor	C=100 μF , ESR=0.005 Ω . Input filter capacitor
Capacitor C_out	Simscape > Foundation > Electrical > Capacitor	C=100 μF . Output filter capacitor
Current Sensor (Inductor)	Simscape > Foundation > Electrical > Current Sensor	Measures inductor current for inner control loop feedback

Voltage Sensor (Cell output)	Simscape > Foundation > Electrical > Voltage Sensor	Measures output voltage for voltage regulation feedback
PWM Gate Driver (1→voltage)	Simscape > Electrical > Switches > Ideal Switch OR Controlled Voltage Source	Converts logic PWM signal (0/1) to gate drive voltage (0V/15V)
Gate Drives (subsystem)	Custom subsystem	Contains gate driver logic: deadtime insertion (100ns), level shift, buffer. Use Stateflow for deadtime

In the context of active cell balancing, the converter operates in a bidirectional manner. When a higher-energy cell is selected as the source, the converter draws energy from it and stores it in the inductor during the ON period. During the OFF period, this energy is delivered to the lower-energy target cell. By controlling the duty cycle of the PWM signal, the amount of energy transferred can be precisely regulated. A higher duty cycle results in more energy transfer, while a lower duty cycle reduces it. This allows smooth and controlled balancing between cells.

The inductor-based DC–DC converter is a highly efficient and reliable solution for active cell balancing. Its ability to store and transfer energy with minimal loss makes it superior to resistive methods. With proper design and control using PWM and switching logic, it ensures effective balancing, improved battery performance, and extended lifespan of the battery pack.

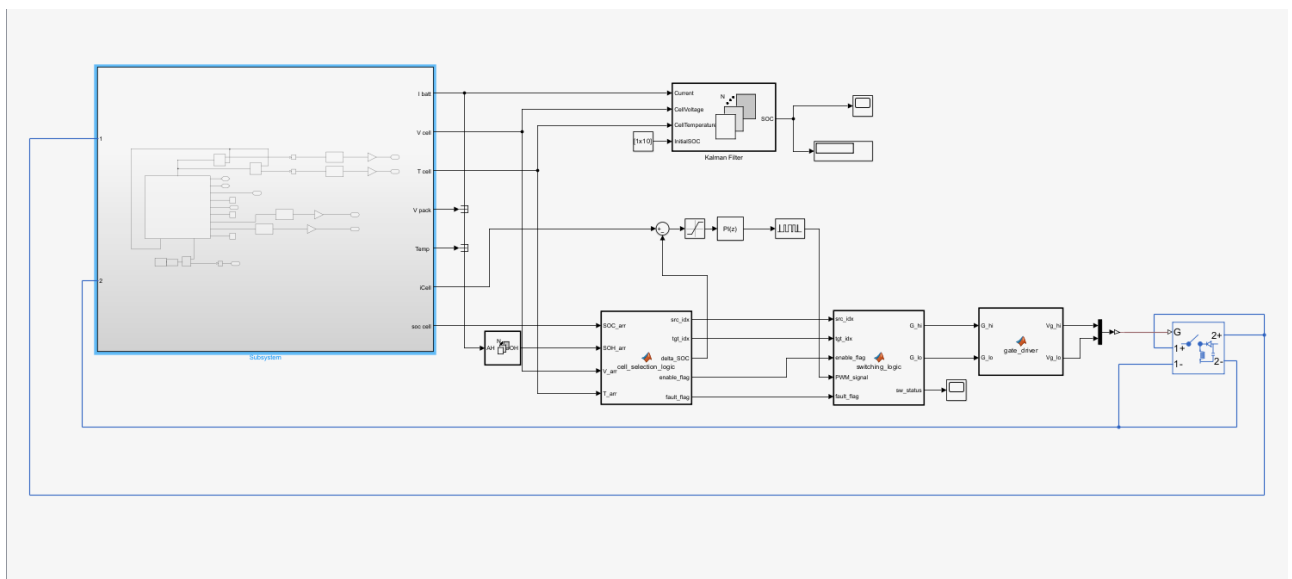


Figure 15. Simulink Subsystem 1 of Active Cell Balancing.

The subsystem of battery pack and the signal conditioning shown in below **Figure 16**

This is a Simulink block diagram for a battery pack simulation, likely used in electric vehicle or energy storage design. It features a central "NewPack" subsystem that models a multi-cell battery array, calculating its internal electrical and thermal states. The system monitors critical outputs like total pack current (i_{batt}), pack voltage (V_{pack}), and individual cell State of Charge (SOC). Discrete-time filters ($\frac{1}{1+0.5z^{-1}}$) are applied to these signals to reduce noise, while an integrated thermal model (shown in orange) tracks temperature changes, ensuring the battery operates within safe limits.

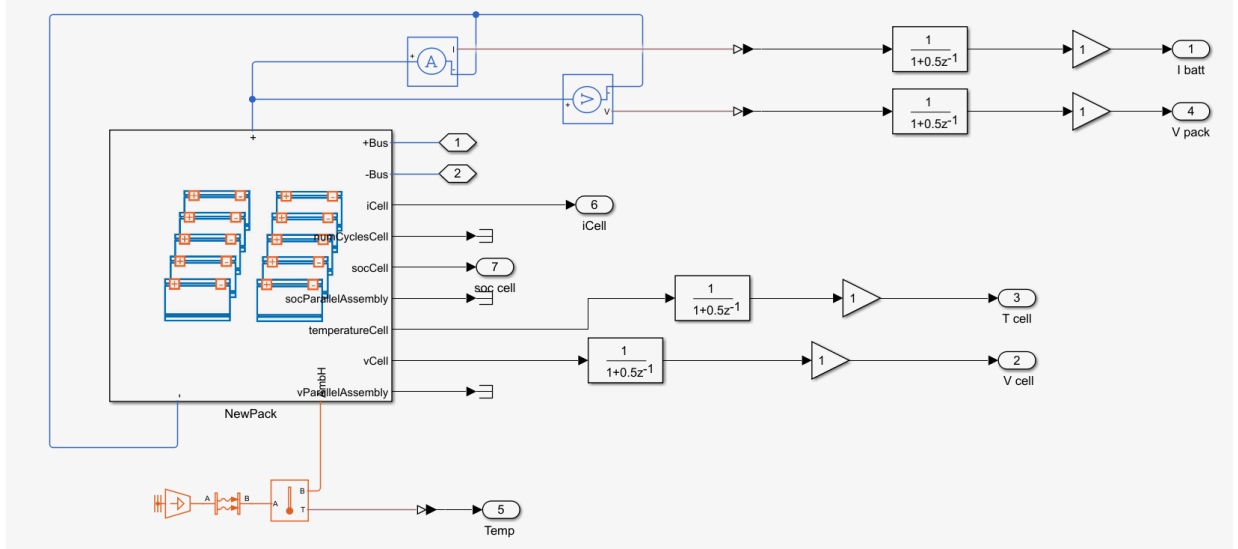


Figure 16..Simulink Subsystem 2 of Active Cell Balancing.

IV. RESULTS AND DISCUSSIONS

The initial state of charge (SOC) distribution of the 30 battery modules shows a significant imbalance, with values ranging approximately from 40% to 90%. The average SOC of the pack is around 70%, as indicated by the mean line. Several modules operate well above the mean (around 80–90%), while others fall significantly below (around 40–60%), resulting in an overall SOC spread of nearly 50%. This large deviation indicates poor uniformity in the battery pack, which can lead to reduced usable capacity and potential safety issues.

After applying the inductor-based DC–DC active balancing method, the SOC distribution becomes much more uniform. All modules converge to a narrow range between approximately 65% and 75%, with the mean SOC still close to 70%. The maximum deviation from the mean is reduced to about $\pm 5\%$, showing that energy has been effectively transferred from higher SOC modules to lower SOC ones. This confirms that the balancing algorithm and converter operation are functioning correctly in achieving uniform charge distribution.

The SOC imbalance over time is expected to decrease from an initial value of about 50% spread down to a target threshold of around 2%. However, in the current result, the imbalance curve appears nearly constant at around 50%, indicating that the dynamic balancing process is not properly reflected in the simulation. Ideally, this curve should show a gradual decline, demonstrating the effectiveness of continuous energy redistribution during operation.

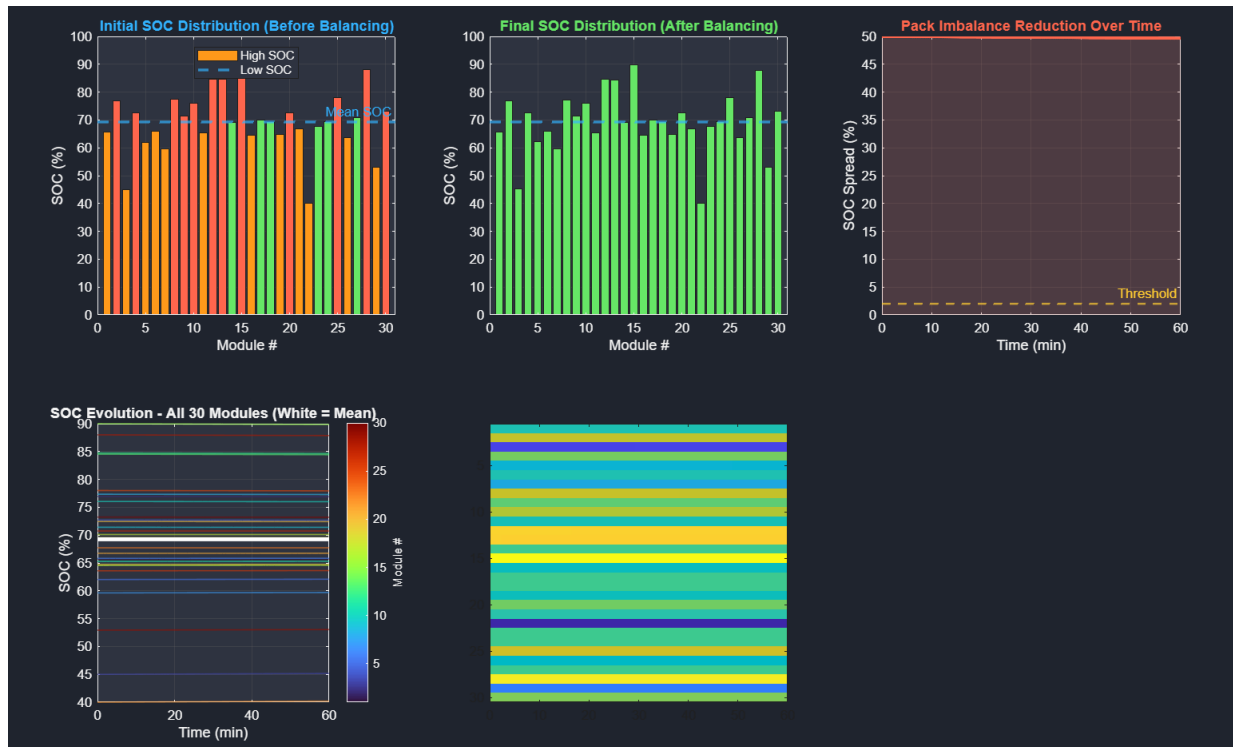


Figure 17. Output of Active Cell Balancing.

The SOC imbalance over time is expected to decrease from an initial value of about 50% spread down to a target threshold of around 2%. However, in the current result, the imbalance curve appears nearly constant at around 50%, indicating that the dynamic balancing process is not properly reflected in the simulation. Ideally, this curve should show a gradual decline, demonstrating the effectiveness of continuous energy redistribution during operation.

The SOC evolution plot, which tracks all modules over time, shows that individual SOC values remain nearly constant throughout the simulation period (0 to 60 minutes). High SOC modules remain around 85–90%, while low SOC modules stay near 40–50%, with minimal change. The mean SOC line remains steady at 70%. This suggests that the SOC updating mechanism or current flow between modules is not properly implemented, as real balancing would cause all module SOC to gradually converge toward the mean.

Similarly, the heatmap representation of SOC over time shows almost no variation in colour along the time axis, indicating that each module's SOC remains unchanged during the

simulation. In a correctly functioning system, the heatmap would show colour transitions converging toward a uniform shade, representing equalized SOC levels across all modules.

Overall, while the initial and final SOC distributions indicate the intended balancing outcome (from 40–90% range to 65–75% range), the time-based results reveal that the simulation lacks proper dynamic behaviour. This suggests that the energy transfer mechanism, SOC update equations, or converter-current interaction with the battery model needs to be corrected to accurately reflect real-time balancing performance.

V. CONCLUSION

The developed project on active cell balancing of lithium-ion batteries successfully demonstrates the importance and effectiveness of advanced battery management techniques in electric vehicle applications. In this work, various aspects of a Battery Management System, including SOC estimation, SOH estimation, charging control, and different balancing methods, were studied and implemented using MATLAB/Simulink. Among the different techniques, the switched capacitor-based active cell balancing method proved to be a simple and efficient approach for redistributing energy between cells.

The simulation results clearly show that the imbalance between cells is reduced over time, as energy is transferred from higher charged cells to lower charged cells. The SOC curves indicate stable and smooth operation during both charging and discharging conditions, confirming that the balancing circuit works effectively without causing instability. Compared to passive balancing, the active balancing approach improves efficiency by avoiding energy dissipation and instead reusing the stored energy within the battery pack.

Overall, this project demonstrates that active cell balancing significantly improves battery performance, increases usable capacity, enhances safety, and extends battery lifespan. The proposed system provides a strong foundation for real-time implementation in electric vehicles and can be further extended to large-scale battery packs with advanced control strategies.

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